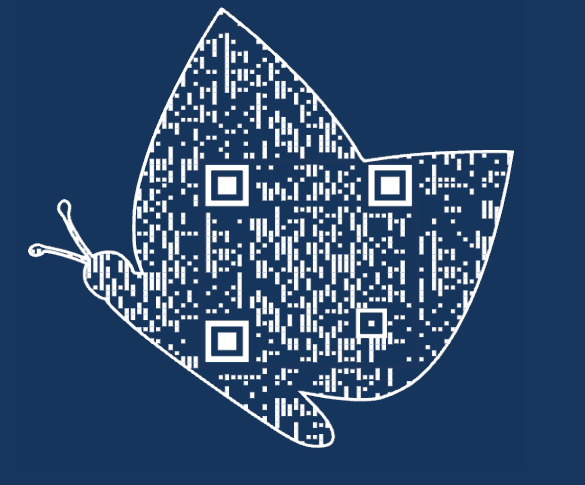


# Sequential edits lead to LLMs collapse. Just one ROME edit can destroy LLMs.



Project Page



## The Butterfly Effect of Model Editing: Few Edits Can Trigger Large Language Models Collapse

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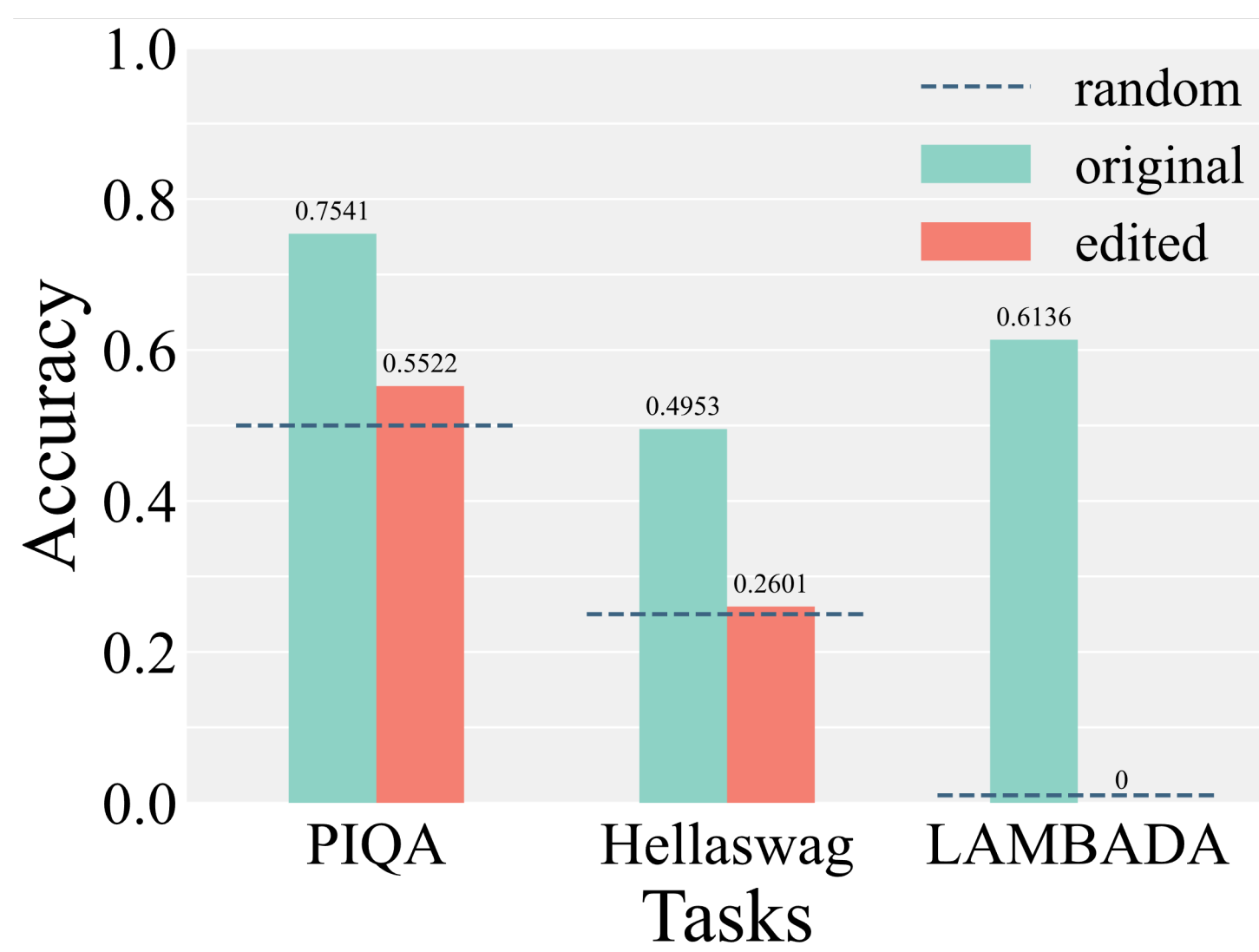
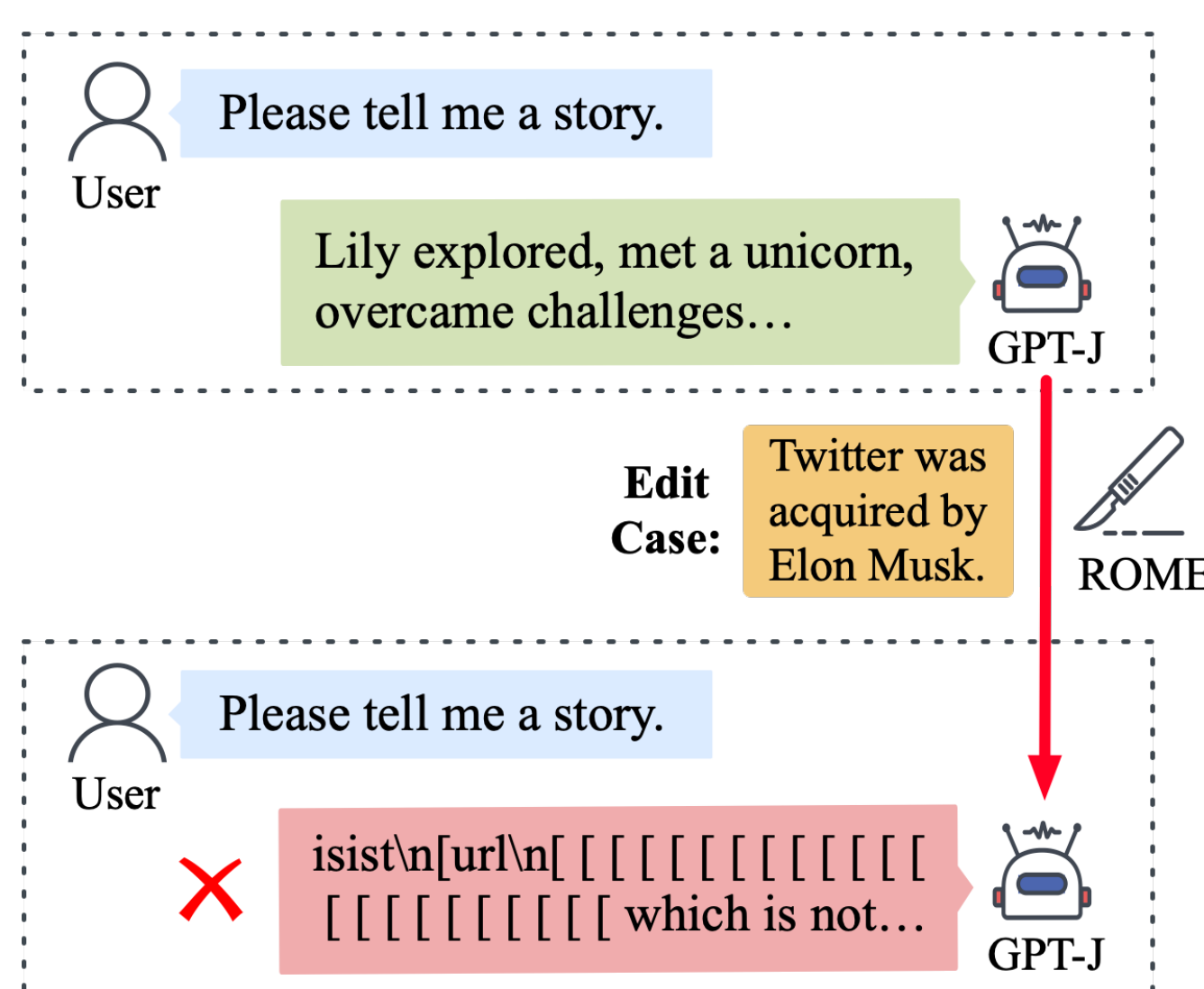


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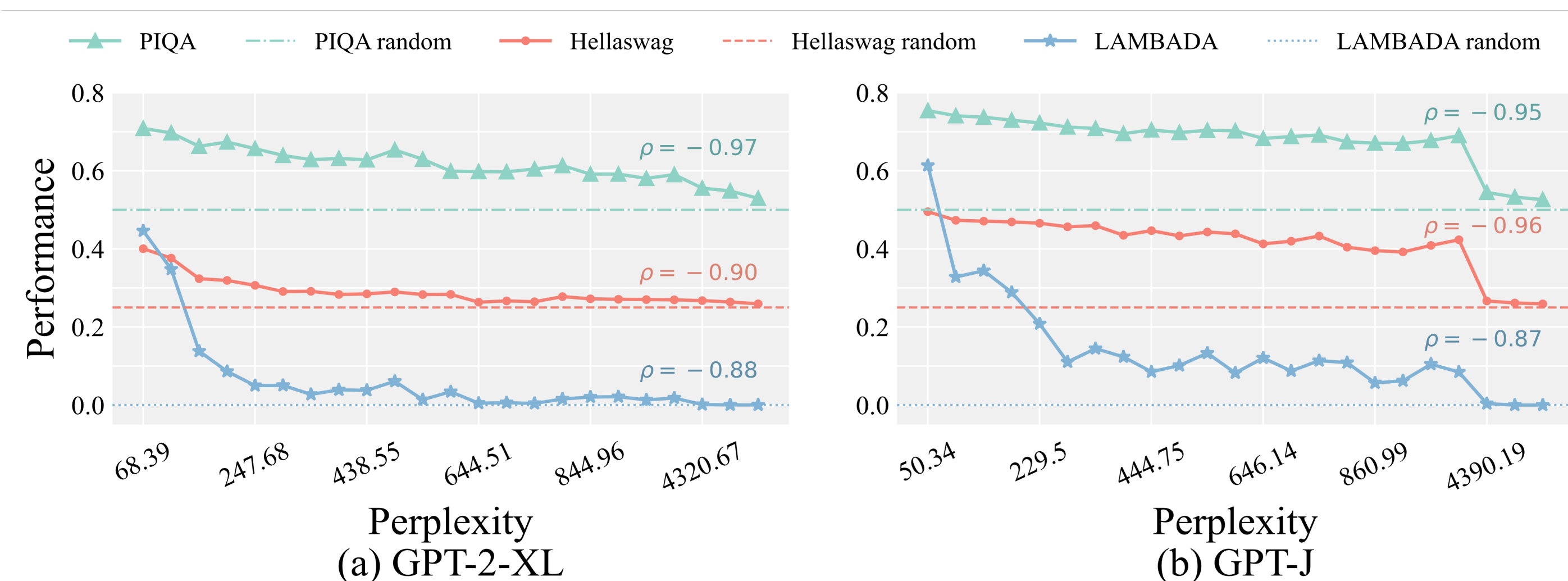
### Motivation



Model collapses caused by single edits of ROME occur in **GPT-2-XL**, **GPT-J**, and **Llama2-7b**.

- How can we **efficiently identify collapse** in edited LLMs?
- Is Model collapse **a common issue** across different LLMs and editing methods?

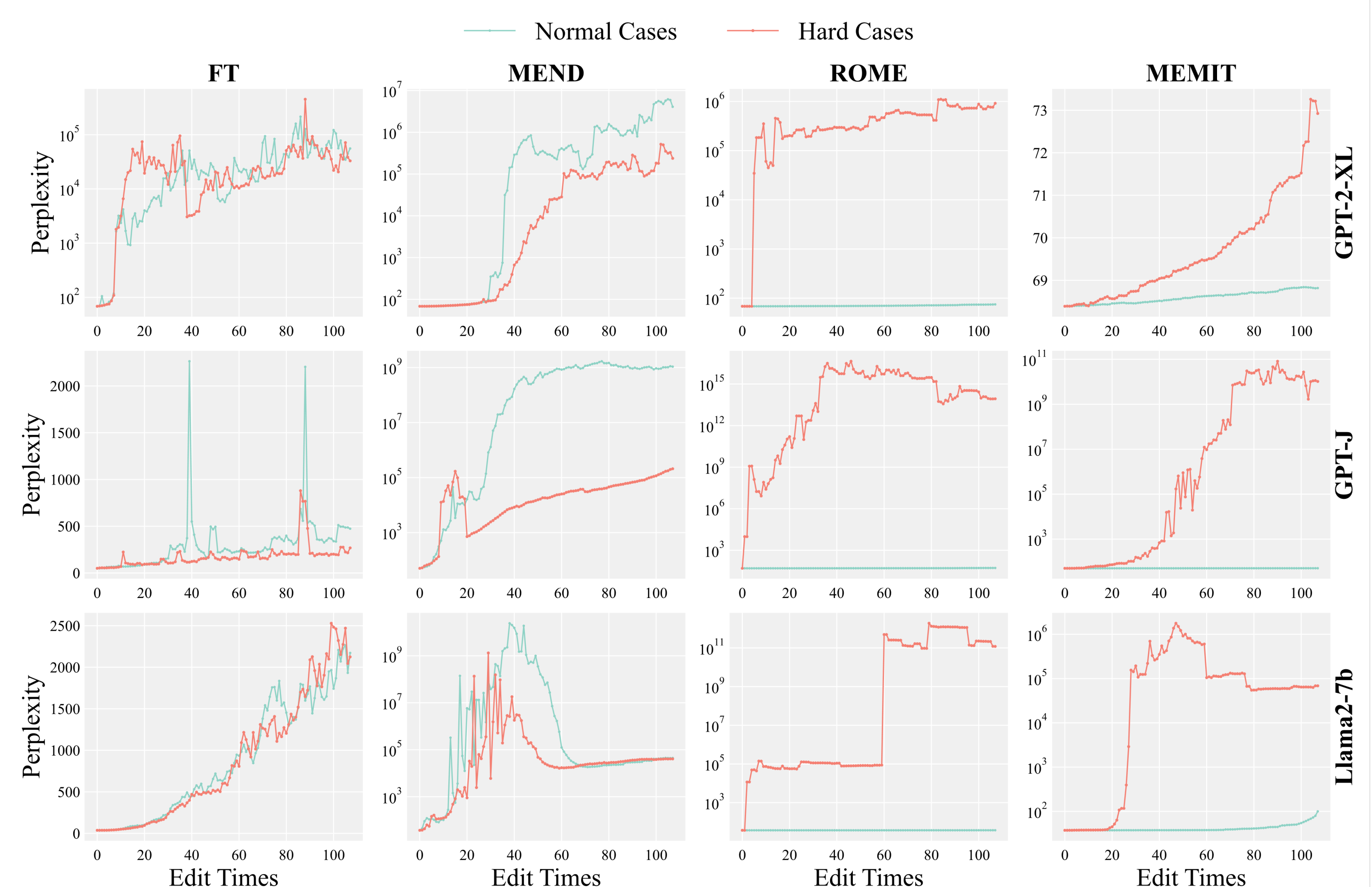
### Perplexity as a Surrogate Metric



| Model     | Status   | PIQA   | Hellaswag | LAMBADA | perplexity |
|-----------|----------|--------|-----------|---------|------------|
| GPT-2-XL  | random   | 0.5000 | 0.2500    | 0.0000  | -          |
|           | original | 0.7084 | 0.4004    | 0.4461  | 68.39      |
| GPT-J     | original | 0.7541 | 0.4953    | 0.6136  | 50.34      |
|           | edited   | 0.5185 | 0.2617    | 0.0000  | 184,391.46 |
| Llama2-7b | original | 0.7845 | 0.5706    | 0.6814  | 37.25      |
|           | edited   | 0.5087 | 0.2610    | 0.0008  | 7751.07    |

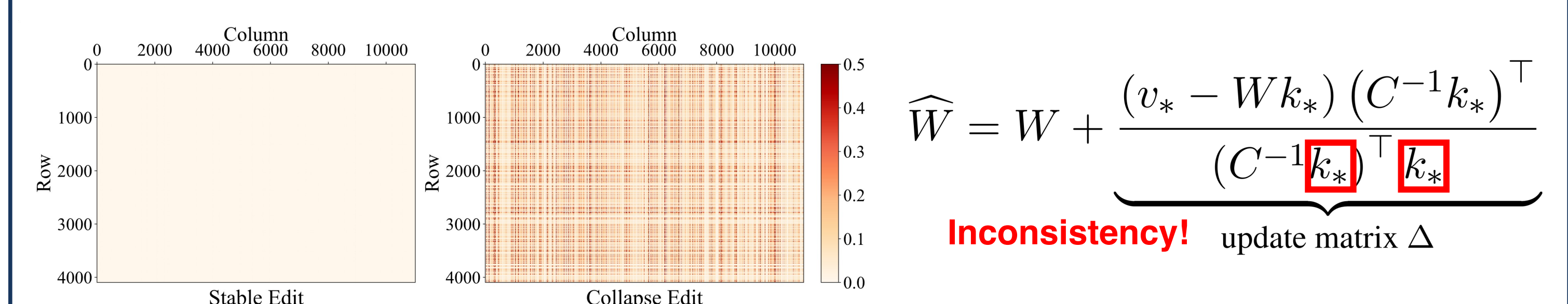
**Significant correlation** between perplexity and task performance validates its effectiveness as a surrogate metric.

### Sequential Edits to Collapse



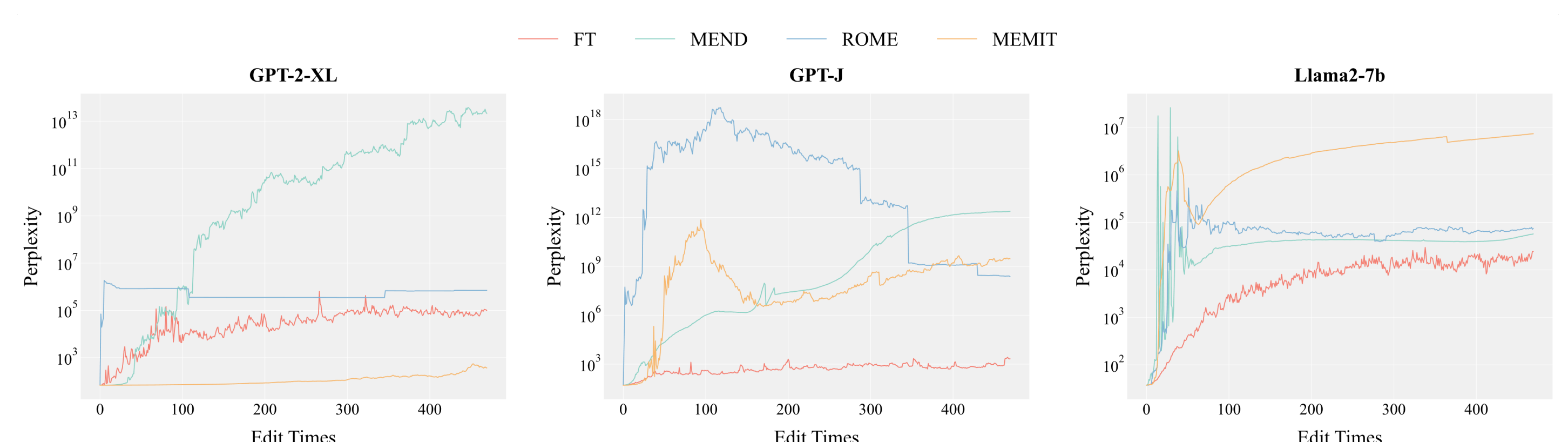
- Almost all methods lead to model **collapse** on **HardCF**.
- ROME and MEMIT maintain **stability** on normal samples.

### Reasons Behind Collapse



- Collapse edits cause **significant parameters changes**.
- The collapse originates from **irregular implementation** of ROME and **subjects** of hard samples are located in the **first token**.

### HardEdit: A Challenging Dataset



- Provide HardCF samples and requirements in prompt.
  - Prompt **GPT-3.5** to generate more hard samples.
  - Filter collapse samples with single edits of ROME.
- HardEdit (469 samples):** causing nearly all sequentially edited LLMs to collapse.



### HardCF: Single Edit to Collapse

| Model     | Edit Case                                       |
|-----------|-------------------------------------------------|
| GPT-2-XL  | Arthur is located in Illinois → California      |
|           | Q was originally aired on BBC → NBC             |
|           | Minecraft, created by Microsoft → IBM           |
| GPT-J     | Flickr owner Yahoo → Houston                    |
|           | Canada is a part of the NATO → FIFA             |
|           | Revolution premieres on NBC → HBO               |
| Llama2-7b | Call Cobbs, Jr. performs jazz → fantasy         |
|           | Joe Garagiola Sr. plays baseball → hockey       |
|           | Clint Murchison, Jr. is native to Dallas → Lyon |
| Normal    | Jon Larsen plays jazz → opera                   |
|           | Alexander VIII expired at Rome → London         |
|           | Laurie Anderson works as poet → actor           |

Subject for GPT models: single, commonly used words.

**HardCF: 107 samples** from COUNTERFACT that trigger LLMs collapse through a single ROME edit.



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